

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Logic Reasoning & Mapping Exercise

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO3: *Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning.* (BTL 3)

Assessment Tool Weightage: 15%

Total Marks: 100

Time: 60 Mins

No. of Questions: 5

Annexure A

Logic Reasoning & Mapping Exercise

Code of Conduct:

I (V Sunil Kumar / 192425292) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V Sunil Kumar

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Knowledge Identification	15%	
2. Logical Representation and Mapping	20%	
3. Predicates, Variables, Constants and Quantifiers	15%	
4. Logical Connectives and Formula Construction	10%	
5. Inference and Logical Reasoning	15%	
6. Unification and Substitution	10%	
7. Resolution and Proof Construction	10%	
8. Mapping Diagram / Clarity and Presentation	5%	
Total	100%	

ANSWER ANY ONE OF THE FOLLOWING

SMART HEALTHCARE DIAGNOSIS SYSTEM

Q1. Smart Healthcare Diagnosis

Problem Statement

A hospital uses an AI-based system to identify possible diseases from patient symptoms. The system receives information about patients and applies logical rules to infer possible diseases and required medical action.

For the given case:

- Ravi has fever.
- Ravi has cough.
- Ravi has body pain.
- Patients with fever and cough may have flu.
- Patients with flu and body pain may have viral infection.
- Patients with viral infection require medical attention.

The objective is to represent this knowledge using Propositional Logic and First-Order Logic and use unification and resolution to determine whether Ravi has a viral infection.

1. Facts, Entities, Predicates, Variables and Constants

1.1 Entity

The main entity in the problem is:

Ravi – the patient being diagnosed.

1.2 Facts

The given facts are:

1. Ravi has fever.
2. Ravi has cough.
3. Ravi has body pain.

In logical form:

```
[  
Fever(Ravi)  
]  
  
[  
Cough(Ravi)
```

]

[

BodyPain(Ravi)

]

1.3 Predicates

The following predicates are used:

Predicate	Meaning
Fever(x)	x has fever
Cough(x)	x has cough
BodyPain(x)	x has body pain
Flu(x)	x has flu
ViralInfection(x)	x has viral infection
MedicalAttention(x)	x requires medical attention

1.4 Variable

Let:

[

x

]

represent any patient.

1.5 Constant

Let:

[

Ravi

]

be the constant representing the particular patient Ravi.

1.6 Rules

The knowledge contains three main rules:

Rule 1

If a patient has fever and cough, the patient may have flu.

[

$Fever(x) \wedge Cough(x) \rightarrow Flu(x)$

]

Rule 2

If a patient has flu and body pain, the patient may have viral infection.

[
 $\text{Flu}(x) \wedge \text{BodyPain}(x) \rightarrow \text{ViralInfection}(x)$
]

Rule 3

If a patient has viral infection, the patient requires medical attention.

[
 $\text{ViralInfection}(x) \rightarrow \text{MedicalAttention}(x)$
]

2. Propositional Logic Representation

In Propositional Logic, each statement is represented by a proposition.

Let:

- **F** = Ravi has fever.
- **C** = Ravi has cough.
- **B** = Ravi has body pain.
- **FL** = Ravi has flu.
- **V** = Ravi has viral infection.
- **M** = Ravi requires medical attention.

2.1 Facts

[
F
]

[
C
]

[
B
]

2.2 Rules

Fever and cough imply flu

[
 $(F \wedge C) \rightarrow \text{FL}$
]

Flu and body pain imply viral infection

[
 $(FL \wedge B) \rightarrow V$
]

Viral infection implies medical attention

[
 $V \rightarrow M$
]

2.3 Complete Propositional Knowledge

[
 $F, \neg C, \neg B$
]

[
 $(F \wedge C) \rightarrow FL$
]

[
 $(FL \wedge B) \rightarrow V$
]

[
 $V \rightarrow M$
]

Therefore:

[
 $F \wedge C \rightarrow FL$
]

[
 $FL \wedge B \rightarrow V$
]

[
 $V \rightarrow M$
]

3. First-Order Logic Representation

First-Order Logic allows us to represent the general rules for any patient.

3.1 Facts

For Ravi:

[
Fever(Ravi)

]

[
Cough(Ravi)

]

[
BodyPain(Ravi)

]

3.2 FOL Rules

Rule 1

For every patient, if the patient has fever and cough, then the patient may have flu.

[
 $\boxed{\forall x[(\text{Fever}(x) \wedge \text{Cough}(x)) \rightarrow \text{Flu}(x)]}$
]

Rule 2

For every patient, if the patient has flu and body pain, then the patient may have viral infection.

[
 $\boxed{\forall x[(\text{Flu}(x) \wedge \text{BodyPain}(x)) \rightarrow \text{ViralInfection}(x)]}$
]

Rule 3

For every patient, if the patient has viral infection, then the patient requires medical attention.

[
 $\boxed{\forall x[\text{ViralInfection}(x) \rightarrow \text{MedicalAttention}(x)]}$
]

3.3 Quantifier Used

The universal quantifier:

[
 $\forall x$
]

means "**for every patient x.**"

No existential quantifier is necessary for the given rules because the problem already identifies Ravi as the specific patient.

4. Logical Connectives and Formula Construction

The following logical connectives are used:

Symbol	Meaning	Example
(\land)	AND	$\text{Fever}(x) \wedge \text{Cough}(x)$
(\rightarrow)	Implication	$\text{Fever}(x) \rightarrow \text{Flu}(x)$
(\neg)	NOT	$\neg \text{Flu}(x)$
(\lor)	OR	$\text{Fever}(x) \vee \text{Cough}(x)$

The main formulas are:

Formula 1

[
 $\text{Fever}(x) \wedge \text{Cough}(x) \rightarrow \text{Flu}(x)$
]

Formula 2

[
 $\text{Flu}(x) \wedge \text{BodyPain}(x) \rightarrow \text{ViralInfection}(x)$
]

Formula 3

[
 $\text{ViralInfection}(x) \rightarrow \text{MedicalAttention}(x)$
]

The complete logical relationship is:

[
 $(\text{Fever}(x) \wedge \text{Cough}(x))$
 $\rightarrow \text{Flu}(x)$
]

[
 $\text{Flu}(x) \wedge \text{BodyPain}(x)$
 $\rightarrow \text{ViralInfection}(x)$
]

[
 $\text{ViralInfection}(x)$
 $\rightarrow \text{MedicalAttention}(x)$
]

5. Unification for Two Relevant Predicates

Unification finds substitutions that make two logical expressions identical.

Unification 1

Consider:

[
Fever(x)
]

and:

[
Fever(Ravi)
]

To make both predicates identical, substitute:

[
x=Ravi
]

Therefore:

[
 $\boxed{\{x/Ravi\}}$
]

is the substitution.

After substitution:

[
Fever(Ravi)=Fever(Ravi)
]

Hence, the predicates successfully unify.

MGU

[
 $\boxed{MGU=\{x/Ravi\}}$
]

Unification 2

Consider:

[
Flu(x)

]

and:

[

Flu(Ravi)

]

Substitute:

[

x=Ravi

]

Therefore:

[

$\boxed{\{x/Ravi\}}$

]

After substitution:

[

Flu(Ravi)=Flu(Ravi)

]

Hence, the predicates successfully unify.

MGU

[

$\boxed{MGU=\{x/Ravi\}}$

]

6. Resolution to Prove Ravi Has Viral Infection

We select the query:

[

$\boxed{ViralInfection(Ravi)}$

]

To prove this using resolution, we assume the opposite of the query:

[

$\neg ViralInfection(Ravi)$

]

If this assumption produces a contradiction with the knowledge base, then the original query is proved.

Step 1: Convert Rule 1 into Clause Form

Original rule:

$$[\text{Fever}(x) \wedge \text{Cough}(x) \rightarrow \text{Flu}(x)]$$

Using:

$$[A \wedge B \rightarrow C]$$

we obtain:

$$[\neg A \vee \neg B \vee C]$$

Therefore:

$$[\neg \text{Fever}(x) \vee \neg \text{Cough}(x) \vee \text{Flu}(x)]$$

For Ravi:

$$[\neg \text{Fever}(\text{Ravi}) \vee \neg \text{Cough}(\text{Ravi}) \vee \text{Flu}(\text{Ravi})]$$

Step 2: Convert Rule 2 into Clause Form

Original rule:

$$[\text{Flu}(x) \wedge \text{BodyPain}(x) \rightarrow \text{ViralInfection}(x)]$$

Clause form:

$$[\neg \text{Flu}(x) \vee \neg \text{BodyPain}(x) \vee \text{ViralInfection}(x)]$$

For Ravi:

$$[\neg \text{Flu}(\text{Ravi}) \vee \neg \text{BodyPain}(\text{Ravi}) \vee \text{ViralInfection}(\text{Ravi})]$$

Step 3: Use Known Facts

Known facts:

[
Fever(Ravi)
]
[
Cough(Ravi)
]
[
BodyPain(Ravi)
]

Step 4: Derive Flu(Ravi)

Clause:

[
 $\neg \text{Fever(Ravi)} \vee \neg \text{Cough(Ravi)} \vee \text{Flu(Ravi)}$
]

Given:

[
Fever(Ravi)
]

Resolving gives:

[
 $\neg \text{Cough(Ravi)} \vee \text{Flu(Ravi)}$
]

Given:

[
Cough(Ravi)
]

Resolving again gives:

[
 $\boxed{\text{Flu(Ravi)}}$
]

Step 5: Derive ViralInfection(Ravi)

Use the clause:

$$\begin{bmatrix} \neg \text{Flu}(\text{Ravi}) \vee \neg \text{BodyPain}(\text{Ravi}) \vee \text{ViralInfection}(\text{Ravi}) \end{bmatrix}$$

We have:

$$\begin{bmatrix} \text{Flu}(\text{Ravi}) \end{bmatrix}$$

and:

$$\begin{bmatrix} \text{BodyPain}(\text{Ravi}) \end{bmatrix}$$

Resolving with both facts gives:

$$\begin{bmatrix} \boxed{\text{ViralInfection}(\text{Ravi})} \end{bmatrix}$$

Step 6: Resolution with Negated Query

We initially assumed:

$$\begin{bmatrix} \neg \text{ViralInfection}(\text{Ravi}) \end{bmatrix}$$

But the knowledge base derives:

$$\begin{bmatrix} \text{ViralInfection}(\text{Ravi}) \end{bmatrix}$$

Therefore:

$$\begin{bmatrix} \text{ViralInfection}(\text{Ravi}) \wedge \neg \text{ViralInfection}(\text{Ravi}) \end{bmatrix}$$

produces the empty clause:

$$\begin{bmatrix} \boxed{\Box} \end{bmatrix}$$

The empty clause represents a contradiction.

Therefore, the negated query is false and the original query is true.

[
 $\boxed{\text{ViralInfection(Ravi)}}$
]

7. Complete Inference Sequence

The complete reasoning process is:

Initial Facts

[
Fever(Ravi)
]
[
Cough(Ravi)
]
[
BodyPain(Ravi)
]

Inference 1

Using:

[
 $\text{Fever}(x) \wedge \text{Cough}(x) \rightarrow \text{Flu}(x)$
]

and substituting:

[
 $x = \text{Ravi}$
]

we obtain:

[
 $\boxed{\text{Flu(Ravi)}}$
]

Inference 2

Using:

[
 $\text{Flu}(x) \wedge \text{BodyPain}(x) \rightarrow \text{ViralInfection}(x)$

]

and substituting:

[

x=Ravi

]

we obtain:

[

\boxed{ViralInfection(Ravi)}

]

Inference 3

Using:

[

ViralInfection(x)\rightarrow MedicalAttention(x)

]

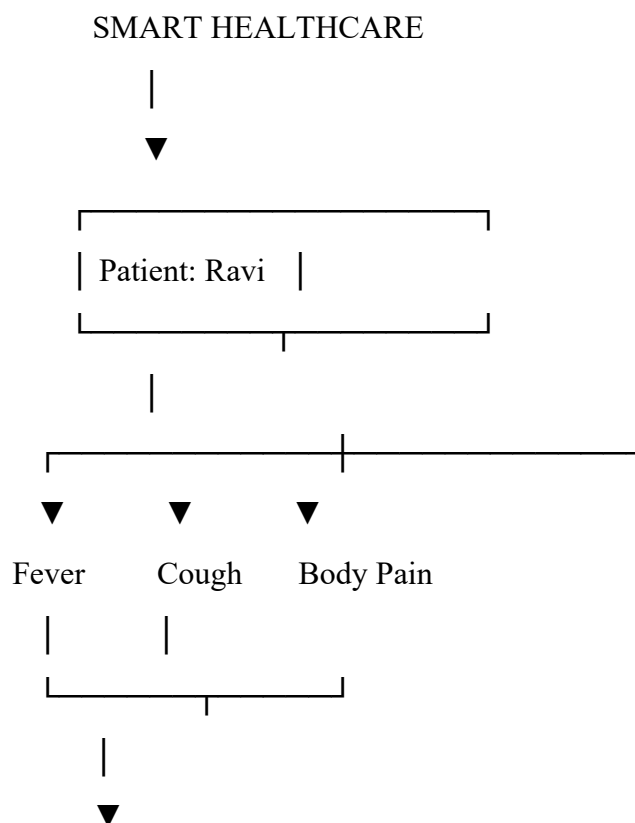
we obtain:

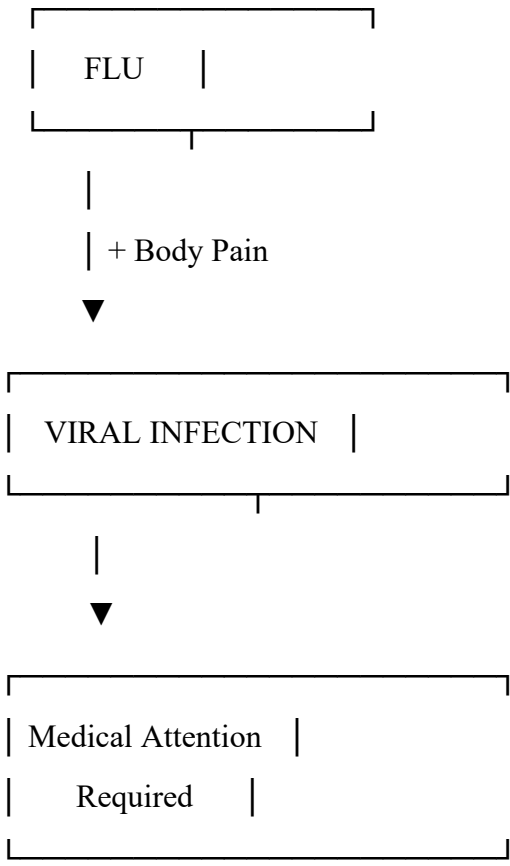
[

\boxed{MedicalAttention(Ravi)}

]

8. Mapping / Logic Diagram





Logical Mapping

```
[
Fever(Ravi)+Cough(Ravi)
]
[
\Downarrow
]
[
Flu(Ravi)
]
[
+\ BodyPain(Ravi)
]
[
\Downarrow
]
[
ViralInfection(Ravi)
]
```

```
[
\Downarrow
]

[
MedicalAttention(Ravi)
]
```

9. Final Conclusion

The AI healthcare system successfully identifies a possible disease from Ravi's symptoms using logical reasoning.

The initial facts are:

```
[
Fever(Ravi), Cough(Ravi), BodyPain(Ravi)
]
```

From fever and cough:

```
[
Fever(Ravi)\land Cough(Ravi)
\rightarrow Flu(Ravi)
]
```

From flu and body pain:

```
[
Flu(Ravi)\land BodyPain(Ravi)
\rightarrow ViralInfection(Ravi)
]
```

Finally:

```
[
ViralInfection(Ravi)
\rightarrow MedicalAttention(Ravi)
]
```

Therefore, the final inference is:

```
[
\boxed{\text{Ravi has a possible viral infection and requires medical attention.}}
]
```

The resolution proof confirms that:

```
[
```


$\boxed{\text{ViralInfection(Ravi)}}$

]

is logically entailed by the given facts and rules.

Final Inference Chain

[

$\boxed{\{$

Fever(Ravi)+Cough(Ravi)

$\rightarrow$ Flu(Ravi)

$\rightarrow$ ViralInfection(Ravi)

$\rightarrow$ MedicalAttention(Ravi)

}

]

This answer covers the required six tasks of Question 1 and the major assessment criteria: knowledge identification, Propositional Logic, FOL, predicates and quantifiers, logical connectives, unification and MGU, resolution proof, inference sequence, and mapping/diagram presentation.

EVALUATION RUBRICS:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
Problem Understanding & Knowledge Identification	15%	13–15% – Clearly identifies entities, facts, relationships, assumptions, and reasoning requirements from the given scenario.	10–12% – Identifies most relevant knowledge with minor omissions.	7–9% – Identifies basic knowledge but misses several important elements.	0–6% – Poor or incorrect identification of the given knowledge.
Logical Representation & Mapping	20%	17–20% – Accurately maps real-world statements into appropriate propositional and first-order logic representations.	13–16% – Provides mostly correct logical mappings with minor errors.	9–12% – Provides partial mappings with several logical errors.	0–8% – Logical mapping is largely incorrect or missing.
Predicates, Variables, Constants & Quantifiers	15%	13–15% – Correctly identifies and applies predicates, variables, constants, universal and existential quantifiers.	10–12% – Uses the concepts correctly with minor errors.	7–9% – Demonstrates partial understanding with some incorrect usage.	0–6% – Concepts are incorrectly applied or missing.
Logical Connectives & Formula Construction	10%	9–10% – Correctly applies AND, OR, NOT, implication, and equivalence to	7–8% – Mostly correct use of connectives with minor errors.	5–6% – Basic use of connectives but several errors occur.	0–4% – Incorrect or incomplete formula construction.

		construct valid logical formulas.			
Inference & Logical Reasoning	15%	13–15% – Performs accurate step-by-step reasoning and derives valid conclusions from the given knowledge.	10–12% – Reasoning is mostly correct with minor gaps.	7–9% – Provides partial reasoning with some incorrect conclusions.	0–6% – Reasoning is incorrect, unsupported, or missing.
Unification & Substitution	10%	9–10% – Correctly identifies substitutions and Most General Unifier (MGU) and applies them accurately.	7–8% – Correctly performs most unification steps with minor errors.	5–6% – Demonstrates basic understanding but has several errors.	0–4% – Unable to perform unification correctly.
Resolution / Proof Construction	10%	9–10% – Correctly applies resolution and presents a clear, logically valid proof of the query.	7–8% – Resolution is mostly correct with minor errors.	5–6% – Shows partial understanding but proof contains gaps.	0–4% – Resolution is incorrect or not demonstrated.
Clarity, Mapping Diagram & Presentation	5%	5% – Provides a clear, well-organized mapping/logic diagram with accurate notation and excellent presentation.	4% – Presentation is clear with minor formatting or diagram issues.	3% – Basic presentation with limited clarity.	0–2% – Poorly organized, unclear, or incomplete presentation.